The World According to the Hereford Mappa Mundi

A medieval map of the world set against a modern one:
the 1,025 names of Bevan and Phillott's index, identified, classified and measured

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Abstract

The Hereford Mappa Mundi (about 1300) is the largest surviving medieval map of the world. I use the index of names compiled by Bevan and Phillott in 1873 as an inventory of the map (1,025 names), assign each to the four kinds of material those authors distinguished, and compare the positions of 164 inscriptions, located by eye on Konrad Miller's 1896 facsimile, with the real places the 1873 book identifies. Of the 985 names that are not part of the map's frame or natural history, 72% are classical, 9% Biblical, 12% medieval and 6% legendary; every Biblical name is in Asia, and 92% of the names in the British Isles are medieval. Britain and Ireland, 0.4% of the land of Africa, Asia and Europe, carry 8.1% of the names. Measured from Jerusalem, the map's centre, directions are kept well once east is put at the top: the median error for the 140 located real places is 19 degrees and 81% lie within 45 degrees. Distances are kept in order (rank correlation 0.90) but compressed towards the rim: places a median 3,747 km away are only 4.8 times as far out on the map as places a median 128 km away. Relative to the map as a whole, the Holy Land is drawn 4.5 times larger, the British Isles 1.5 times, and India and Central Asia at 0.71 of the average. The map spends its space on Scripture and on home.

Keywords: medieval cartography; mappa mundi; Hereford; historical geography; map distortion; Pleiades; open data

1 Introduction

The Hereford Mappa Mundi, drawn on a single sheet of vellum about 158 by 133 cm and kept in Hereford Cathedral, is the largest medieval map of the world known to survive (Wikipedia, 2026). It follows the T and O pattern: the inhabited world is a disc ringed by ocean, with east at the top, Jerusalem at the centre, Asia filling the upper half and Europe and Africa the lower quarters, divided by the Mediterranean, the Don and the Nile. It is a religious and encyclopedic object rather than a guide for travellers, and it mixes Biblical history, classical geography, the marvels of the Alexander romance and a detailed picture of the mapmaker's own island.

Bevan and Phillott's *Mediaeval Geography* (Bevan and Phillott, 1873) went over the map name by name, region by region, identifying each name where they could and tracing it to its source, and ended with an index of names. That index, and the scanned text of the book, are the core of this study. I ask four questions. How many names does the map carry, and in which parts of the world? What share is Biblical, classical, medieval and legendary? Which real places did it include? And how distorted is it, in direction, in distance and in the space it gives to each region? The paper accompanies an interactive comparison of the map and a modern map (https://maps.tristanhotham.com/mappa-mundi/).

2 Data

The book. The Internet Archive's scan of Bevan and Phillott (1873) (public domain) provides the text, divided here into the book's 174 pages using the running heads. The book's chapters follow the map: the frame and the ocean (chapter I), Asia (II to IV), Africa (V), the Mediterranean and Black Sea with their islands (VI), Europe (VII and VIII) and the British Isles (IX). The map's own words are printed in black-letter type, which the OCR renders as nonsense; the commentary, identifications and citations are in roman type and read cleanly. Miller's *Die Herefordkarte* (Miller, 1903) was used for cross-checks.

The images. Konrad Miller's 1896 facsimile (Miller, 1896), a hand-drawn copy with clear labels (3,840 by 4,708 pixels as used), and a photograph of the original (Wikimedia Commons, an enlargement of a 600 pixel image, so its labels cannot be read). Both are in the public domain.

Gazetteers. Pleiades, a gazetteer of ancient places (Pleiades, 2026), version 4.1, and Natural Earth countries and towns (Natural Earth, 2026). Facts on the map's size, date and contents are from English Wikipedia (Wikipedia, 2026).

3 Methods

3.1 Inventory

The book's index, "Index of the names of places and objects [on the] Hereford Mappa Mundi", has 1,025 entries once lines broken by the scan are rejoined, each with a type where given (river, island, mountain, sea, promontory, gulf) and the page or pages that discuss it. I corrected OCR errors in the index names (rn read as m, N as K, and so on) and in ten page numbers, checking each against the text. The part of the world is that of the chapter the first page falls in. The index records names, not pictures: a town with a label is one entry, an unlabelled picture is none, and some legends are indexed under their key word.

3.2 Classes

Bevan and Phillott sort the map's materials into Biblical, classical, legendary and contemporaneous (medieval) (Bevan and Phillott, 1873, Introduction, section 11). I applied that scheme with a fixed rule, in order. (1) The winds, quarters and continent labels of the frame and the real animals and plants (40 names) are left out. (2) Legendary: monstrous peoples, fabulous beasts, and the places and objects of legend and myth, from a list drawn up name by name. (3) Biblical: every name in the book's sections on the rivers of Paradise (pp. 25 to 26), Palestine and Arabia (pp. 77 to 81) and the Exodus in Egypt (pp. 85 to 86), less the Greek and Roman names there, and a few Biblical names elsewhere (Babel, Babylon, Ur, Nineveh, Noah's Ark, Enos, Ophir). (4) Medieval: in the British Isles every name except five from Roman writers; in the rest of Europe a list of post-Roman places, regions and peoples; in Asia and Africa only six names of medieval origin (Samarkand, the Turks, the monasteries of St Antony, St Zosimas and Mahun). (5) Classical: everything else. The lists are in the published code.

3.3 Real places

For each inscription I located on the facsimile (below) I read the book's identification and recorded it with the page and the book's words, and gave the modern place's coordinates, checked against Natural Earth. For the other names I looked for the same name in Pleiades (attested and romanised names and place titles), allowing simple medieval spellings (y for i, f for ph, ablative endings), within a box for the part of the world the book discusses the name under; approximate spellings were allowed with a similarity of at least 0.86 and the same first two letters. Every match was checked by hand against the part of the world the book puts the name in and the place it points to; 78 that pointed to another place of the same or a similar name were rejected. In all 337 of the 928 names that are neither frame nor legend (36%) have a real position: 143 from the book and 194 from Pleiades (32 of them with rough Pleiades locations). This is a lower bound on what can be identified.

3.4 Positions on the map

No coordinate dataset exists for the Hereford map, so I located 164 well-known and legible inscriptions by eye on the facsimile, at the town picture or, where there is none, at the middle of the label, recording my confidence (119 read clearly, 45 with some doubt because the label is worn, crowded or ambiguous). Of these, 16 are legendary and 143 are identified with a real place (140 distinct places once Babel and Babylon, and Santiago and its shrine, are merged). Positions were carried to the photograph by an affine fit on 9 features clear on both images (Jerusalem, Paradise, the Pillars of Hercules, the Tower of Babel, Delos, Crete, Charybdis, the Well of the Sun and Lake Salinarum), with a root mean square error of 44 pixels, about 1.4% of the map's width.

3.5 Distortion

For each located real place I measured the true great-circle distance and initial bearing from Jerusalem (31.78 N, 35.23 E) and the distance and bearing on the facsimile from the centre of the walled circle of Jerusalem, reading east as up. Map distance is expressed as a share of the map radius, the mean distance from Jerusalem to Paradise and to the Pillars of Hercules (1,536 pixels). To summarise the scale of each region, I took all pairs of located places in the region at least 100 km apart and computed the median of map distance over true distance, divided by the same median over all such pairs on the map; regions with fewer than five located places are not scored. Finally, I fitted the similarity transform (rotation, uniform scale and shift, least squares) that best lays the located map positions on an azimuthal equidistant projection centred on Jerusalem, and measured each place's displacement from where that fit puts it.

4 Results

4.1 What is on the map

Table 1 and Figure 1 show the inventory. Asia has 371 names, Europe 324, Africa 117, the seas and islands 105 and the British Isles 83. 640 are settlements, regions or peoples, 139 rivers, 108 islands and 69 mountains. For comparison, Wikipedia counts about 420 towns, 15 Biblical events, 33 animals and plants, 32 people and 5 scenes from classical mythology (Wikipedia, 2026).

By the rule above, 714 names (72%) are classical, 93 (9%) Biblical, 121 (12%) medieval and 57

Table 1: Names in Bevan and Phillott's index by part of the map and class (frame and natural history left out).

Part of the map	Names	Biblical	Classical	Medieval	Legendary
Asia	360	93	225	6	36
Africa	114	0	103	0	11
Europe	321	0	279	39	3
British Isles	83	0	5	76	2
Seas and islands	105	0	100	0	5
All	985	93	714	121	57

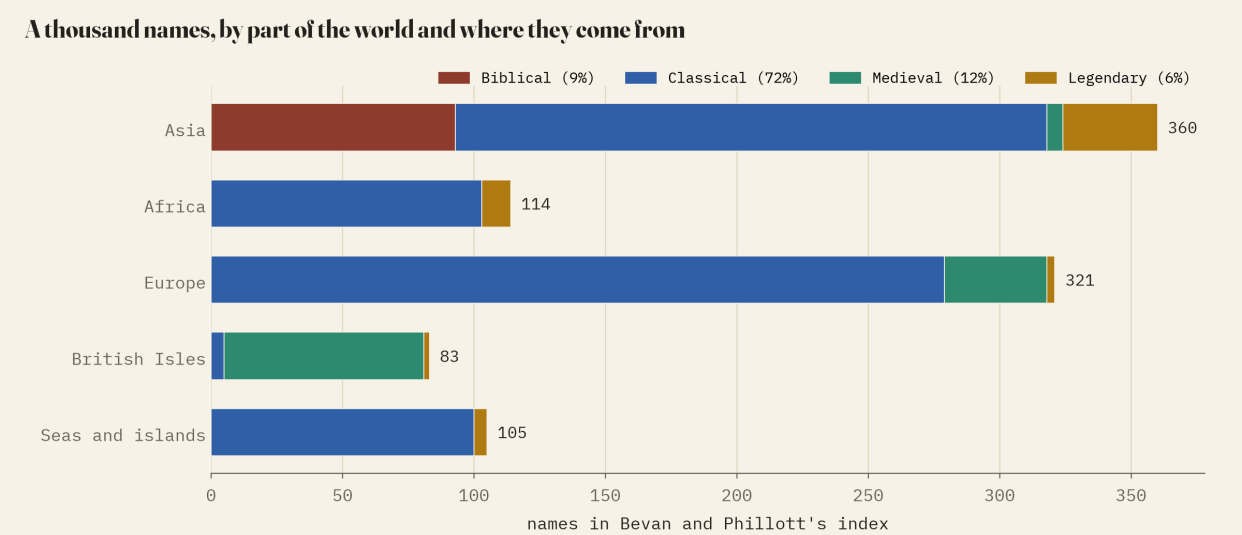


Figure 1: Names in Bevan and Phillott's index by part of the map and class.

(6%) legendary. All 93 Biblical names are in Asia, as are 36 of the legendary ones. Africa's names are 90% classical, Europe's 87%. The British Isles are the exception, 92% medieval, consistent with Bevan and Phillott's remark that the mapmaker's contemporary knowledge is "almost entirely confined to Europe" and with the detail of the British section.

4.2 Real places and their positions

Figure 2 shows the located inscriptions on the facsimile and the identified places on a modern map; Appendix Table 3 lists them.

4.3 Direction

Bearings from Jerusalem are kept well (Figure 3). The median absolute error is 19 degrees, 81% of places are within 45 degrees and all are within 90. The errors are not random: North Africa (69 degrees) and Egypt (72) are swung southward, because the Mediterranean is drawn as a straight channel from the Pillars of Hercules at the bottom of the map to the Holy Land at its centre, while the British Isles (8), Mesopotamia and Persia (12) and India and Central Asia (10) point close to the right way. The best-fitting rotation turns the map's top to a bearing of 98 degrees, a little south of due east, and explains 0.84 of the variance of the true positions; the median displacement from the fitted position is 516 km (Figure 6).

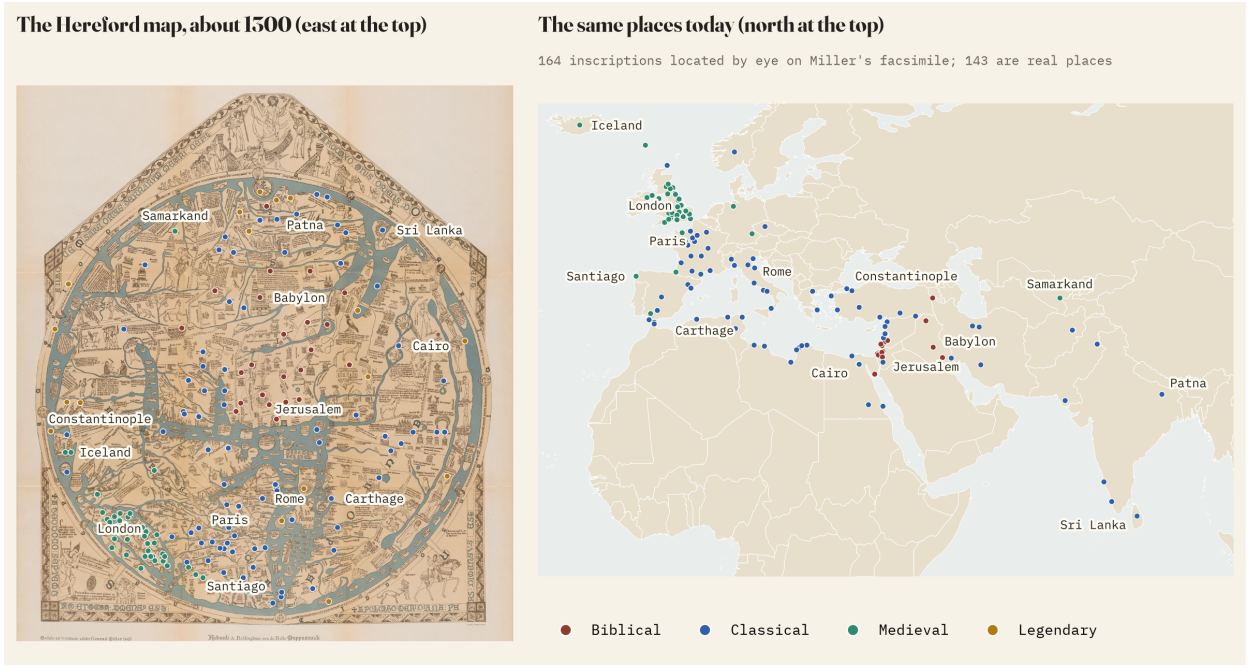


Figure 2: Located inscriptions on Miller's facsimile (left, east at the top) and the places the 1873 book identifies, on a modern map (right). Colours are classes.

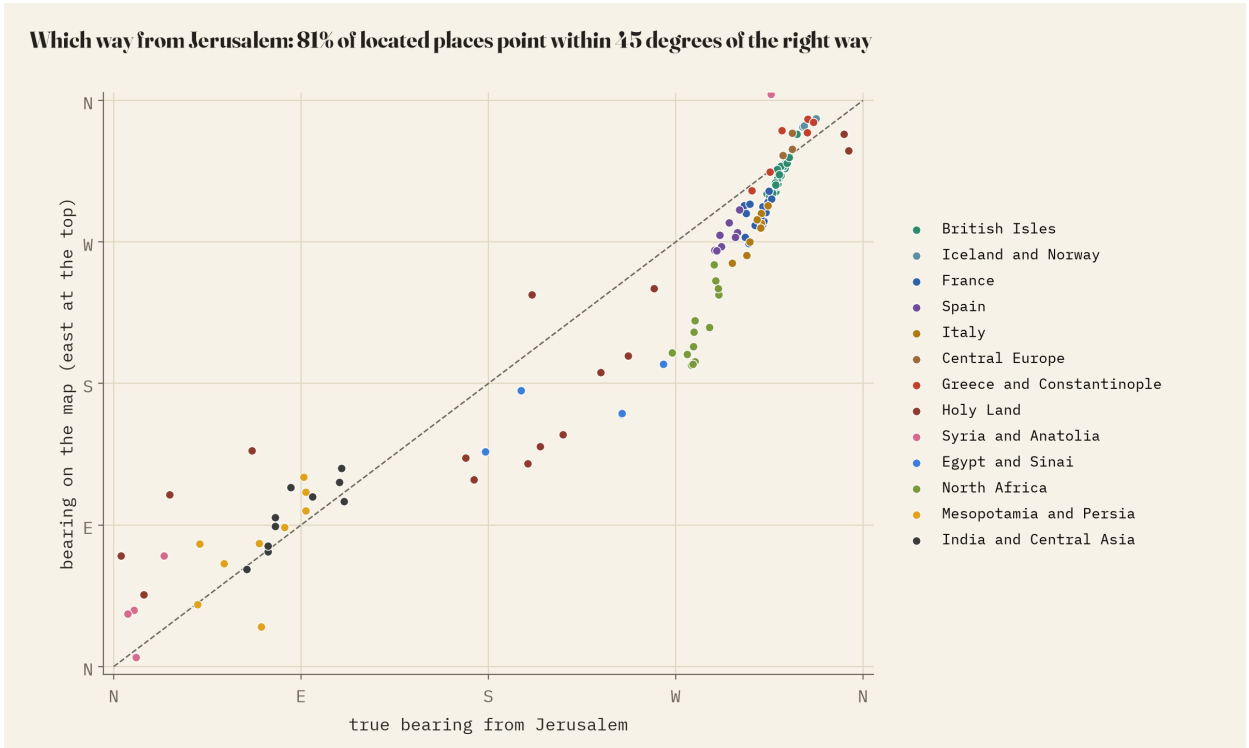


Figure 3: Bearing from Jerusalem on the map (east up) against the true bearing, by region. The dashed line is perfect agreement.

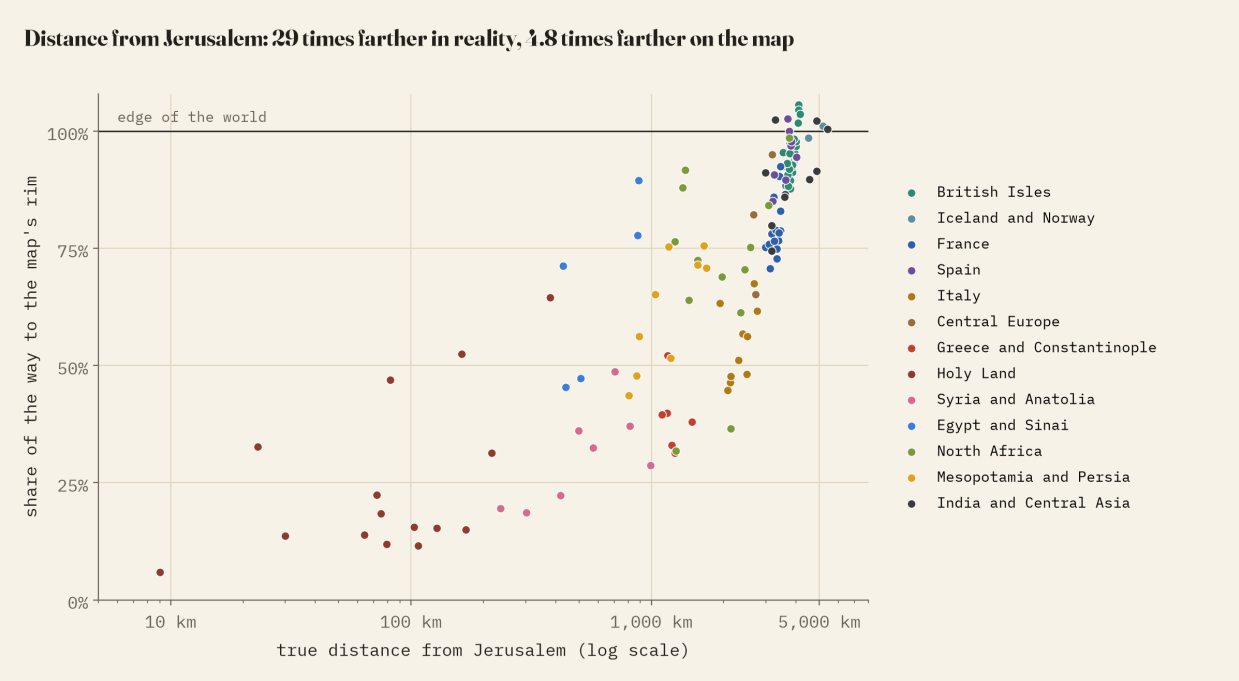


Figure 4: Distance from Jerusalem on the map, as a share of the way to the rim, against true distance (log scale).

4.4 Distance

Distances keep their order (Spearman's rank correlation 0.90) but not their scale (Figure 4). Places within 500 km of Jerusalem (median 128 km) sit a median 20% of the way to the rim; places 3,000 km or more away (median 3,747 km) sit at 93%: 29.3 times as far in reality, 4.8 times on the map. Bethlehem, 9 km from Jerusalem, is 6% of the way out; London, 401 times farther, is at 90%; Sri Lanka, 5,398 km away, sits on the rim.

4.5 Scale by region

Table 2 and Figure 5 give the relative scale of each region. The Holy Land is drawn at 4.5 times the map's average, Egypt and Sinai at 1.9, Mesopotamia and Persia at 1.6, Syria and Anatolia at 1.5 and the British Isles at 1.5; Spain at 0.90 and India and Central Asia at 0.71. Iceland, the Faroes and Norway, too few to score, are squeezed into the rim.

4.6 The British Isles

The British Isles have 83 names, 8.1% of the index, against 0.4% of the land of Africa, Asia and Europe in Natural Earth's outlines, about 21 times their share. Bevan and Phillott judged them "drawn on a scale out of all proportion to the other parts of the map" and the rivers of the Humber depicted with "a degree of accuracy which implies personal acquaintance"; they also noted gross errors, above all Caen, transferred "by a strange blunder" from Normandy to Devon (Bevan and Phillott, 1873, pp. 163 to 170). Figure 7 compares the section with a modern map.

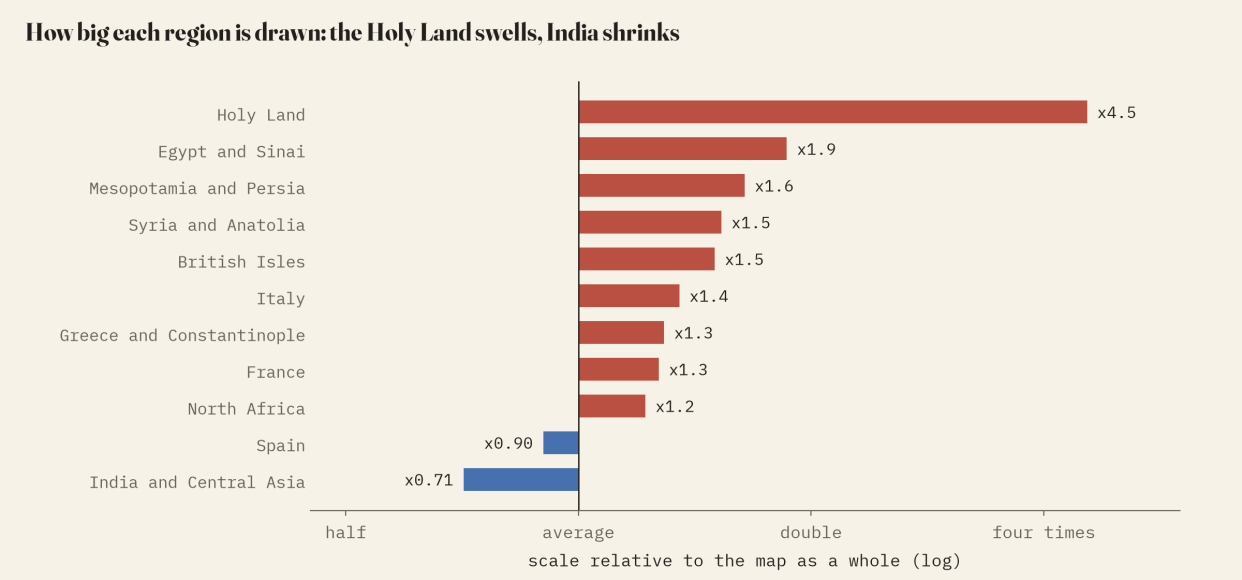


Figure 5: Scale of each region relative to the map as a whole (median ratio of map to true distance over pairs of located places).

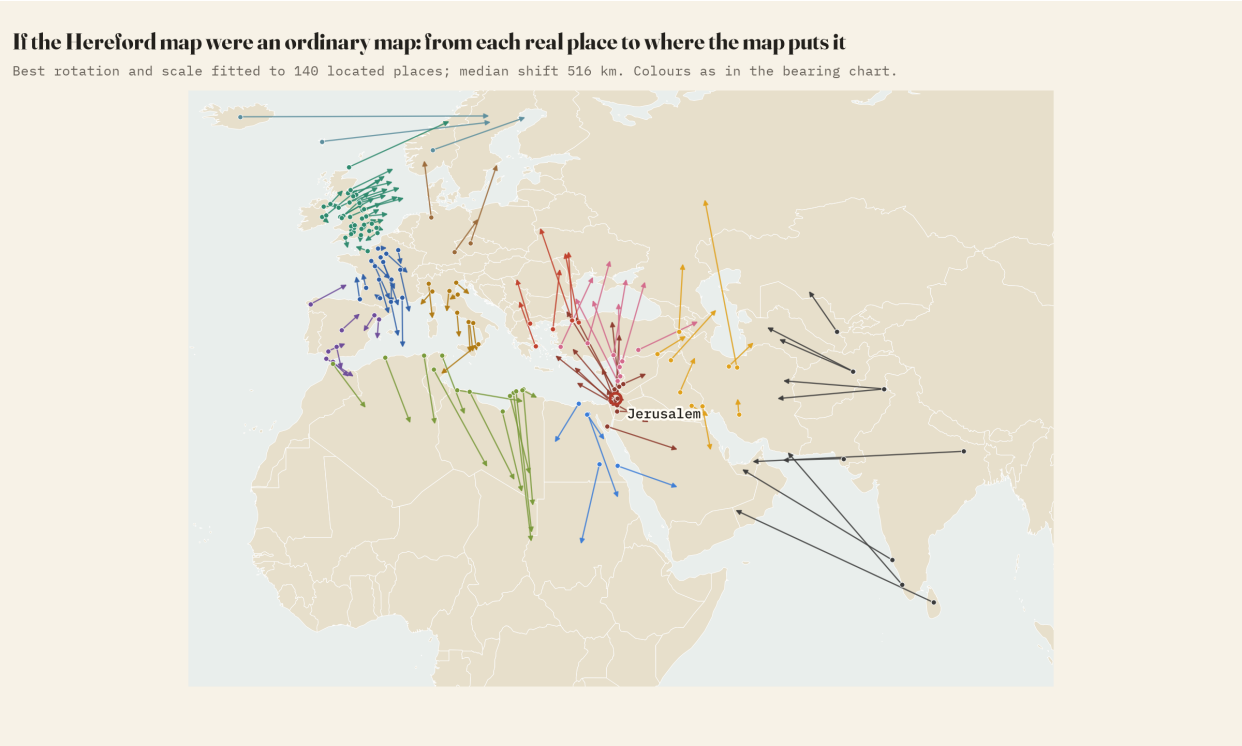


Figure 6: Displacement of each located place from where the best-fitting rotation and scaling of the map would put it.

Table 2: Located real places by region: number, median true distance from Jerusalem, median distance on the map as a share of the radius, median absolute bearing error (degrees), relative scale, and median displacement from the best-fitting position (km).

Region	Places	km	Map	Error	Scale	Shift
British Isles	35	3,801	96%	8	1.50	249
Iceland and Norway	3	4,498	99%	12	too few	1,325
France	15	3,325	78%	24	1.27	384
Spain	8	3,717	96%	24	0.90	344
Italy	10	2,352	54%	34	1.35	388
Central Europe	3	2,720	82%	3	too few	804
Greece and Constantinople	6	1,193	39%	8	1.29	884
Holy Land	15	82	16%	50	4.55	776
Syria and Anatolia	8	536	31%	28	1.53	1,062
Egypt and Sinai	5	508	71%	72	1.86	973
North Africa	13	1,970	72%	69	1.22	1,102
Mesopotamia and Persia	9	1,183	65%	12	1.64	534
India and Central Asia	10	3,594	90%	10	0.71	1,434

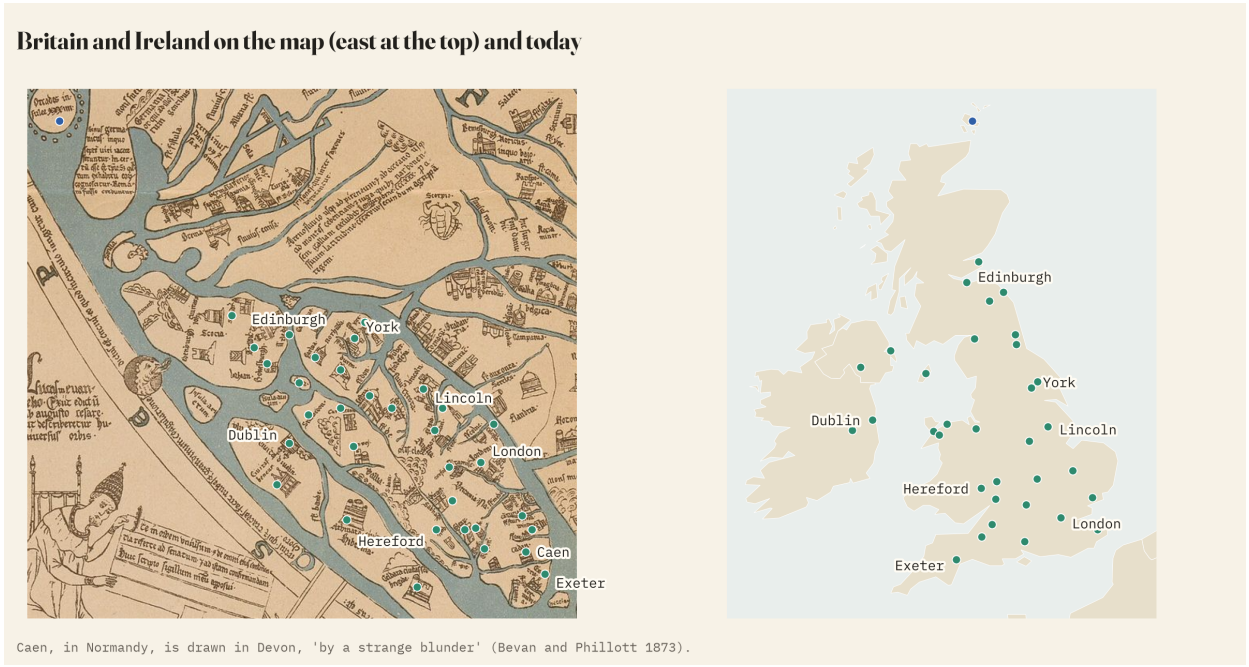


Figure 7: Britain and Ireland on the facsimile (east at the top) and today.

5 Limitations

First, the inventory is Bevan and Phillott's index, which records names, not pictures, and is not a complete modern edition; Westrem's edition and the Virtual Mappa project transcribe every inscription. Second, the classes rest on judgements made name by name, under a stated rule; others would draw some boundaries differently, especially between classical and medieval names in Europe. Third, the 1873 identifications are the Victorians' and some have been revised since; Pleiades matches by name are a lower bound on what can be identified. Fourth, positions were located by eye on a nineteenth century copy; the located set favours legible, well-known places, so the distortion figures describe the map's real towns rather than its whole surface. Fifth, the photograph is low resolution and the affine fit to it is approximate.

6 Conclusion

Measured against a modern map, the Hereford Mappa Mundi is not a failed attempt at geography but a different kind of map. Its directions from Jerusalem are mostly right once east is at the top, and its distances keep their order; what it gives up is scale, spending its space on the Holy Land, Egypt and Mesopotamia, where the Bible's history happened, and on the British Isles, where its makers lived, and pressing the rest of the world against the encircling ocean. Most of its names are inherited from late Roman books; the medieval present appears mainly at home.

Data and code availability

The interactive comparison is at https://maps.tristanhotham.com/mappa-mundi/. Data and code are at https://github.com/Tristan-Hotham/mappa-mundi and archived with this paper at https://doi.org/10.5281/zenodo.23878612. Derived data are released under CC BY 4.0 and code under the MIT licence. The book and the map images are in the public domain; Pleiades data are CC BY 3.0.

AI statement

I wrote the text, with drafting and editing assistance from Claude (Anthropic), which also helped prepare the analysis code. I reviewed and checked all text and results.

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A The located inscriptions

Table 3: The 164 inscriptions located on Miller's facsimile, with the 1873 book's page, the class, the identification and my confidence in the position.

Name (index)	Page	Class	Identification (Bevan and Phillott)	Position
Apollonia	92	Classical	Apollonia (Susa, Libya)	high
Aree Philenoram	92	Classical	the Altars of the Philaeni	high
Arsinoe	92	Classical	Arsinoe (Tocra, Teuchira)	high
Berenice	92	Classical	Benghazi (Berenice)	high
Blemee	103	Legendary	(no real place)	medium
Calpel	99	Classical	Gibraltar (Calpe)	high
Cartago	95	Classical	Carthage	high
Cesarea	99	Classical	Caesarea in Mauretania (Cherchell)	medium
Cirene	92	Classical	Cyrene (Shahhat)	high
Fortunate	106	Legendary	(no real place)	medium
Ippone	97	Classical	Hippo Regius (Annaba)	high
Leptis Magna	93	Classical	Leptis Magna	high
Ptholomaida	92	Classical	Ptolemais (Tolmeita)	high
Sabrata	93	Classical	Sabratha	medium
Scinopodes	102	Legendary	(no real place)	high
Septem Montes	99	Classical	Ceuta	high
Suffetula	96	Classical	Sufetula (Sbeitla)	high
Alexandria	88	Classical	Alexandria	high
Alexandria	36	Classical	Alexandria in the Caucasus (Bagram)	medium
Antiochia	68	Classical	Antioch in Pisidia	medium
Antiochia	76	Classical	Antioch of Syria (Antakya)	high
Arbor Sicca	25	Legendary	(no real place)	medium
Area Noe	69	Biblical	the mountains of Armenia (Ararat)	high
Aree Alexandri	28	Classical	(no real place)	high
Ascalon	78	Biblical	Ashkelon	high
Avalerion	30	Legendary	(no real place)	high
Babel Tttkris	74	Biblical	Babylon (the Tower of Babel)	high
Babylonia	74	Biblical	Babylon	medium
Babylonia	86	Classical	Cairo	high
Berenice	84	Classical	Berenice on the Red Sea	medium
Beritus	76	Classical	Beirut	high
Bersabee	80	Biblical	Beersheba	high
Bethleem	79	Biblical	Bethlehem	high
Bucephala	28	Classical	Bucephala on the Hydaspes (Jhelum)	medium
Calcidonia	69	Classical	Chalcedon (Kadikoy)	medium
Carax	72	Classical	Charax Spasinou	high
Carimaspi	61	Legendary	(no real place)	medium
Carmelum	78	Biblical	Mount Carmel	medium
Caspia	52	Classical	(no real place)	high
Caspie Portee	64	Classical	the Caspian Gates	high
Cassica	30	Classical	Capissa (Kapisa)	medium
Cesarea Philippi	78	Biblical	Caesarea Philippi (Banias)	high
Cottonare	27	Classical	Kochi (Cochin)	medium
Damascus	77	Biblical	Damascus	high
Ebron	80	Biblical	Hebron	medium
Enos	27	Biblical	(no real place)	high
Ephesus	68	Classical	Ephesus	high
Gaza	79	Biblical	Gaza	high
Gerara	79	Biblical	Gerar	high
Gigantes	27	Legendary	(no real place)	high
Griphe	61	Legendary	(no real place)	high
Hur	74	Biblical	Ur of the Chaldees	high
Jerico	80	Biblical	Jericho	medium

Name (index)	Page	Class	Identification	Position
Jerusalem	79	Biblical	Jerusalem	high
Laodiciam	76	Classical	Latakia (Laodicea of Syria)	high
Loth uxor	77	Biblical	(no real place)	high
Magog	49	Legendary	(no real place)	medium
Mandragora	87	Frame	(no real place)	high
Nazaret	78	Biblical	Nazareth	high
Nicea	28	Classical	Nicaea on the Hydaspes (Jhelum)	medium
Nicomedia	69	Classical	Nicomedia (Izmit)	medium
Ninevee	73	Biblical	Nineveh (Mosul)	high
Nisibi	74	Classical	Nisibis (Nusaybin)	high
Orrea Josepbi	86	Biblical	the Pyramids of Giza	high
Paradisi porte	25	Legendary	(no real place)	high
Patalus	28	Classical	the Indus delta (Patale)	medium
Persepolis	72	Classical	Persepolis	high
Petra	77	Classical	Petra	medium
Phenix	85	Legendary	(no real place)	high
Pigmei	28	Legendary	(no real place)	high
Polibrota	32	Classical	Patna (Palimbothra)	high
Rages	71	Classical	Rhagae (Rey, Tehran)	high
Samarcan	52	Medieval	Samarkand	high
Samosaka	73	Classical	Samosata	high
Siene	84	Classical	Aswan (Syene)	high
Sinay	81	Biblical	Mount Sinai	high
Sodom	77	Biblical	Sodom, by the Dead Sea	medium
Tapbana	34	Classical	Sri Lanka (Taprobane, Ceylon)	high
Tarsus	68	Classical	Tarsus	high
Tholomaida	78	Biblical	Acre (Ptolemais)	high
Tripolis	76	Classical	Tripoli, Lebanon	high
Troja	69	Classical	Troy	medium
Zimarim	33	Classical	Muziris, now probably Mangalore	high
Arhmata	172	Medieval	Armagh	high
Bathe	170	Medieval	Bath	medium
Bencur	172	Medieval	Bangor, County Down	medium
Berwic	171	Medieval	Berwick-upon-Tweed	high
Cadan	170	Medieval	Caen (misplaced in England)	high
Canturia	170	Medieval	Canterbury	medium
Carlua	169	Medieval	Carlisle	high
Carnarvan	171	Medieval	Caernarfon	high
Castello Novo	169	Medieval	Newcastle upon Tyne	medium
Celdara	172	Medieval	Kildare	high
Cestria	169	Medieval	Chester	high
Colcestria	168	Medieval	Colchester	high
Cunwey	171	Medieval	Conwy	high
Divelin	172	Medieval	Dublin	medium
Durem	169	Medieval	Durham	high
Eboracum	169	Medieval	York	high
Edenburgh	171	Medieval	Edinburgh	medium
Ely	168	Medieval	Ely	high
Excestria	170	Medieval	Exeter	medium
Glearum	169	Medieval	Gloucester	medium
Glestonia	170	Medieval	Glastonbury	medium
Hreford	169	Medieval	Hereford	high
Kircham	169	Medieval	Kirkham Priory	medium
Lincoln	168	Medieval	Lincoln	high
Londonia	168	Medieval	London	high
Man	172	Medieval	the Isle of Man	high
Norhanton	168	Medieval	Northampton	medium
Oxeford	168	Medieval	Oxford	high
Rokesburg	171	Medieval	Roxburgh	high
S. Andr	171	Medieval	St Andrews	high

Name (index)	Page	Class	Identification	Position
Snawedon	171	Medieval	Snowdon	high
Snotingham	169	Medieval	Nottingham	high
Wintona	170	Medieval	Winchester	high
Wircestr	169	Medieval	Worcester	medium
Arelas	148	Classical	Arles	high
Arimmuin	130	Classical	Rimini	medium
Athene	134	Classical	Athens	medium
Augustudunum	145	Classical	Autun	medium
Aurelianum	146	Classical	Orleans	medium
Avernis	147	Classical	Clermont-Ferrand	high
Basella	126	Medieval	Seville	high
Bayona	148	Medieval	Bayonne	high
Bononia	130	Classical	Bologna	high
Braga	152	Classical	Prague	high
Brema	152	Medieval	Bremen	high
Burdegala	147	Classical	Bordeaux	medium
Carnotum	146	Classical	Chartres	high
Cenomani	146	Classical	Le Mans	high
Cinocephales	159	Legendary	(no real place)	high
Compostella	125	Medieval	Santiago de Compostela	high
Constantinopolis	136	Classical	Istanbul (Constantinople)	high
Corduba	126	Classical	Cordoba	high
Fareie	162	Medieval	the Faroe Islands	high
Ganzmir	158	Legendary	(no real place)	high
Genua	131	Classical	Genoa	medium
Iacobi Templum	125	Medieval	Santiago de Compostela (the shrine of St James)	high
Ilerda	126	Classical	Lleida (Lerida)	high
Limogena	147	Classical	Limoges	high
Narbona	143	Classical	Narbonne	high
Neapolis	131	Classical	Naples	high
Noreya	158	Classical	Norway	high
Orcades	161	Classical	Orkney	high
Parisius	145	Classical	Paris	high
Puteolis	131	Classical	Pozzuoli	high
Remesburgh	154	Medieval	Regensburg	high
Remis	145	Classical	Reims	high
Roma	130	Classical	Rome	high
Rotomagum	145	Classical	Rouen	high
Salema	131	Classical	Salerno	high
Terracona	126	Classical	Tarragona	high
Thessalonica	135	Classical	Thessaloniki	high
Toletum	126	Classical	Toledo	high
Tolosa	147	Classical	Toulouse	medium
Turonis	147	Classical	Tours	medium
Vercellis	130	Classical	Vercelli	medium
Ysland	162	Medieval	Iceland	high
Caribdis	116	Legendary	(no real place)	high
Gades	112	Classical	Cadiz	high
Laborintus	116	Legendary	(no real place)	high
Messana	116	Classical	Messina	high
Venicia	118	Classical	Venice	high